Status	Finished
Started	Tuesday, 29 September 2026, 3:19 PM
Completed	Tuesday, 29 September 2026, 3:29 PM
Duration	10 mins 18 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By dividing the list into halves
b. By comparing each element in the list sequentially ✓
c. By using a hash function
d. By sorting the list first

Question **2**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the left sublist ✓
b. In the right sublist
c. At the beginning of the list
d. In the entire list

Question **3**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is somewhere in the middle of the array ✓
b. When the item is the last element in the array
c. When the item is not the array at all
d. Item is the last element in the array or item is not there at all

Question **4**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(n log n)
b. O(1)
c. O👎 ✓
d. O(log n)

Question **5**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It compares elements sequentially
b. It always starts from the beginning of the list
c. It can be applied only if the list is sorted ✓
d. It works on unsorted lists

Question **6**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Sequential search
b. Hash search
c. Linear search
d. Binary search ✓

Question **7**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Compare the target element with the middle element in the list ✓
b. Sort the list
c. Compare the target element with the first element in the list
d. Divide the list into two equal parts

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. Number values should only be present

Number values should only be present

b. The list must be sorted

c. There must be a mechanism to delete and/or insert elements in the list ✓

d. There should be direct access to the middle element in any sublist

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Binary search algorithm is not efficient when the data elements more than 1500 ✓

b. Must use a sorted array

c. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

d. There must be a mechanism to access middle element directly

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Skipping every second element

b. Sorting the list before searching

c. Checking each element sequentially ✓

d. Dividing the list in half repeatedly

Question **11**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 100

b. 89 and 94

c. 94 and 99

d. 90 and 99 ✓

Question **12**

Incorrect

Mark 0.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Hash search
b. Linear search
c. Binary search ✗
d. Both (1) & (3)

Question **13**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search stops immediately ✓
b. The search starts over from the beginning
c. The search continues until the end of the list
d. The search backtracks to find duplicate elements

Question **14**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 90 and 94
b. 90 and 99 ✓
c. 89 and 99
d. 89 and 94

Question **15**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The first element is returned
b. The last element is returned
c. An error is raised
d. The search is considered unsuccessful ✓